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Parametric Deflection Approximations for End-Loaded, Large-Deflection Beams in Compliant Mechanisms

Geometric nonlinearities often complicate the analysis of systems containing large-deflection members. The time and resources required to develop closed-form or numerical solutions have inspired the development of a simple method of approximating the deflection path of end-loaded, large-deflection cantilever beams. The path coordinates are parameterized in a single parameter called the pseudo-rigid-body angle. The approximations are accurate to within 0.5 percent of the closed-form elliptic integral solutions. A physical model is associated with the method, and may be used to simplify complex problems. The method proves to be particularly useful in the analysis and design of compliant mechanisms.

Introduction

Linearized beam equations prove to be adequate for many applications requiring deflection analysis. These equations tend to be less accurate, however, as deflections become large and the small-deflection linearization less applicable. The difference between large- and small-deflection assumptions lies in the treatment of the Bernoulli-Euler beam equation, which states that the bending moment is proportional to beam curvature. Using the exact equation for curvature, the beam equation can be written as

$$M = EI\frac{d\theta}{ds} = EI\frac{\dfrac{d^2y}{dx^2}}{\left(1+\left(\dfrac{dy}{dx}\right)^2\right)^{3/2}} \tag{1}$$

where M is the moment, $d\theta/ds$ the rate of change of angular deflection along the length of the beam, y the transverse deflection, x the coordinate along the undeflected beam axis, and EI the flexural rigidity. For small deflections, the square of the slope, $(dy/dx)^2$, can be assumed to be negligible compared to unity in the denominator of Eq. (1). The curvature may then be approximated by the second derivative of y, i.e., d^2y/dx^2. This linearization leads to the classical beam moment-curvature equation (Roark and Young, 1982):

$$M = EI\frac{d^2y}{dx^2} \tag{2}$$

The above assumption is not valid in the analysis of large-deflection beams since the slope of the transverse deflection is no longer small. An accurate analysis method, which takes into account the resulting nonlinearities, is discussed later.

An important application of large-deflection analysis techniques is found in the analysis and design of compliant mechanisms (Burns and Crossley, 1966; Shoup and McLarnan, 1971; Her, 1986; Nahvi, 1991). A compliant mechanism is one which gains some or all of its motion from the relative flexibility of its members rather than from rigid-body joints only (Howell and Midha, 1993). For example, consider the two simple compliant mechanisms shown in Fig. 1 (Salamon, 1989). If not for the motion allowed by compliance in the flexible members, both mechanisms would be structures and unable to perform prescribed tasks. Typically, the motion of such mechanisms results from deflections that are large, and therefore nonlinear.

There are several methods available that take into account the nonlinearities introduced by large deflections. A classical method, which yields a closed-form solution, involves the solution of a second-order, nonlinear differential equation using elliptic integrals (Bisshopp and Drucker, 1945; Frisch-Fay, 1969; Mattiasson, 1981). The advantage of this solution technique is that it provides closed-form solutions. The disadvantage is that the derivations are cumbersome and solutions exist for only relatively simple geometries and loadings. Numerical methods, such as the finite element method, are capable of solving much more general problems (Yang, 1973; Bathe and Bolourch, 1979; Her, 1986; Nahvi, 1991).

Numerical methods are typically used in the analysis of compliant mechanisms when an initial design has already been obtained. This initial design may be very difficult to obtain, however, since the nonlinearities of large deflections are coupled with the desired kinematic motion of the overall mechanism. Approximating the nonlinear, large-deflection characteristics of flexible members by means of mathematical or physical models would be helpful in the expeditious design

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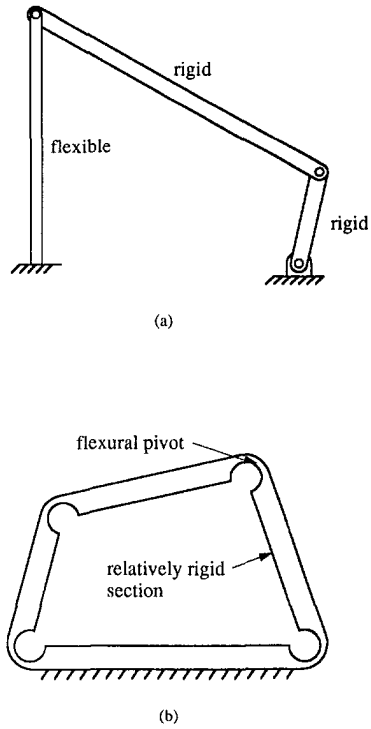

Fig. 1 *(a)* **A partially compliant and** *(b)* **a fully compliant mechanism**

of compliant mechanisms. The development of such a model is the purpose of this paper.

Large-deflection approximations currently exist, which directly approximate the elliptic integral solutions. Frisch-Fay (1963) truncated the infinite series of the complete elliptic integral of the first kind to develop an approximation for the lateral deflection of a post-buckled column. Bisshopp (1973) also used truncated series expressions to derive simplified equations for a cantilever beam with a single transverse load. These equations yield transverse deflections which are valid to within 2 percent error for a load index, $\alpha > 1$. The parametric approximations developed herein differ from the above efforts in that a physical model is used as an aid in their development, rather than directly approximating the elliptic integral equations. This paper develops approximate equations for large-deflection cantilever beams with combined end forces, and no end moments, and expresses them in a single parameter, Θ, termed the pseudo-rigid-body angle.

Large deflection approximations prove to be very useful in compliant mechanisms design, as well as in their motion visualization. Burns and Crossley (1968) approximated the kinematics of flexible-link mechanisms by assuming that large-deflection members transverse a path with a radius of 5/6 the member length. Howell and Midha (1993) developed a method of modeling compliant mechanisms with flexible segments that are short in length compared to the more rigid sections. An example will be given to illustrate the use of the parametric approximations developed herein in compliant mechanism analysis.

Cantilever Beam with End Forces

An initially straight cantilever beam with combined (non-follower) axial and transverse end forces is shown in Fig. 2. The combined end forces may also be treated as a single force acting at an angle ϕ, where $\phi = \tan^{-1}(1/-n)$, as shown. A closed-form solution for the end deflection of the beam may be derived, with the assumption that the beam is linearly elastic, inextensible, rigid in shear, and of constant cross section. The internal moment at any point in the beam is given by

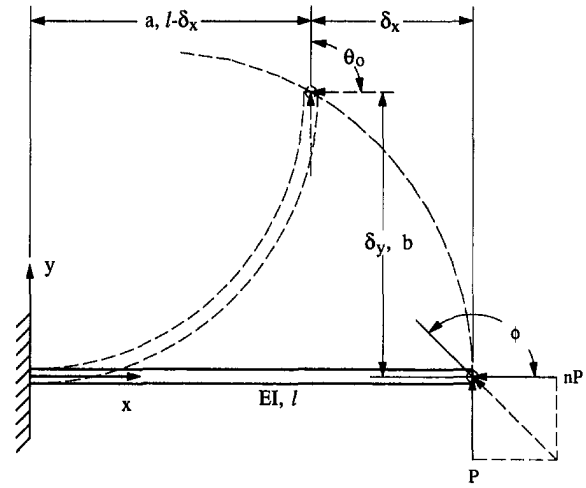

Fig. 2 Cantilever beam with end forces

$$M = P(a-x) + nP(b-y) \tag{3}$$

Using the Bernoulli-Euler equation described above, the curvature may be written as

$$\frac{d\theta}{ds} = \frac{P}{EI}\left[(a-x) + n(b-y)\right] \tag{4}$$

and

$$\frac{d^2\theta}{ds^2} = \frac{P}{EI}\left(-\frac{dx}{ds} - n\frac{dy}{ds}\right)$$

or

$$\frac{d^2\theta}{ds^2} = \frac{-P}{EI}(n \sin\theta + \cos\theta) \tag{5}$$

Integrating yields

$$\frac{1}{2}\left(\frac{d\theta}{ds}\right)^2 = \frac{P}{EI}(n\cos\theta - \sin\theta) + C_1 \tag{6}$$

Using the conditions at the beam end, where $\theta = \theta_o$ and $d\theta/ds = 0$, C_1 is found as

$$C_1 = \frac{P}{EI}(\sin\theta_o - n\cos\theta_o) \tag{7}$$

Substituting Eq. (7) into Eq. (6), and solving for $d\theta/ds$ yields:

$$\frac{d\theta}{ds} = \sqrt{2}\,\frac{\alpha}{l}\sqrt{\lambda - \sin\theta + n\cos\theta} \tag{8}$$

where

$$\alpha^2 = \frac{Pl^2}{EI} \tag{9}$$

$$\lambda = \sin\theta_o - n\cos\theta_o \tag{10a}$$

or

$$\lambda = \eta\cos(\theta_o - \phi_1) \tag{10b}$$

and

$$\eta = \sqrt{1+\eta^2}; \quad \phi_1 = \tan^{-1}\frac{1}{-n} \tag{11}$$

Separating variables in Eq. (8) and integrating yields

$$\frac{\alpha\sqrt{2}}{l}\int_0^l ds = \int_0^{\theta_o}\frac{d\theta}{\sqrt{\lambda - \sin\theta + n\cos\theta}} \tag{12}$$

or (see Byrd and Friedman, 1954, or Howell, 1991, for details of integration):

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for $\theta_o < \phi_1$

$$\alpha = \frac{1}{\sqrt{\eta}}\left[F\left(\frac{\pi}{2}, t\right) - F(\gamma, t)\right] \tag{13}$$

where

$$\gamma = \sin^{-1}\sqrt{\frac{\eta - n}{\eta + \lambda}} \tag{14}$$

$$t = \sqrt{\frac{\eta + \lambda}{2\eta}} \tag{15}$$

and $F(\beta, q)$ is an incomplete elliptic integral of the first kind with amplitude β and modulus q (Hancock, 1958; Greenhill, 1959) defined as

$$F(\beta, q) = \int_0^\beta \frac{d\phi}{\sqrt{1 - q^2\sin^2\phi}}$$

The final vertical position of the end of the beam may now be calculated. The curvature, k, may be rewritten as

$$k = \frac{d\theta}{ds} = \frac{d\theta}{dy}\frac{dy}{ds} = \frac{d\theta}{dy}\sin\theta \tag{16}$$

Substituting Eq. (16) into Eq. (8), separating variables, and rearranging yields:

$$\left(\frac{b}{l}\right) = \frac{\sqrt{2}}{2\alpha}\int_0^{\theta_o} \frac{\sin\theta\, d\theta}{\sqrt{\lambda - \sin\theta + n\cos\theta}} \tag{17}$$

Integrating the above equations results in (see Byrd and Friedman, 1954, or Howell, 1991, for integrating details):

$$\frac{b}{l} = \frac{1}{\alpha\eta^{5/2}}\left\{\eta\left[F\left(\frac{\pi}{2}, t\right) - F(\gamma, t) + 2\left(E(\gamma, t) - E\left(\frac{\pi}{2}, t\right)\right)\right] + \eta\sqrt{2\eta(\eta + \lambda)}\cos\gamma\right\} \tag{18}$$

where $E(\beta, q)$ is an incomplete integral of the second kind with amplitude β and modulus q defined as

$$E(\beta, q) = \int_0^\beta \sqrt{1 - q^2\sin^2\phi}\, d\phi$$

The final horizontal position may be found in a similar manner, only now the curvature is written as

$$k = \frac{d\theta}{ds} = \frac{d\theta}{dx}\frac{dx}{ds} = \frac{d\theta}{dx}\cos\theta \tag{19}$$

Substituting Eq. (19) into Eq. (8), separating variables and integrating yields:

$$\frac{a}{l} = \frac{1}{\alpha\eta^{5/2}}\left\{-n\eta\left(F\left(\frac{\pi}{2}, t\right) - F(\gamma, t) + 2\left(E(\gamma, t) - E\left(\frac{\pi}{2}, t\right)\right)\right) + \sqrt{2\eta(\eta + \lambda)}\cos\gamma\right\} \tag{20}$$

The traditional method of displaying large-deflection results is to plot the beam end coordinates as a function of the load. Valuable insight into the nature of the deflections may be gained by plotting these in an alternative manner. Figure 3 shows the tip deflections plotted in $[(a/l), (b/l)]$ plane for various values of n. These curves represent the beam tip deflection paths which minimize the beam strain energy for given load combinations (Gorski, 1976). A symbol is placed on each curve at every 2 deg. of beam end angular deflection. Figure 4 shows a closer view of the beginning of the deflection paths. Two points are surmised from the data plotted in Figs. 3 and 4. These are:

(*i*) the deflection paths for all loads are remarkably similar at the beginning of the path, and

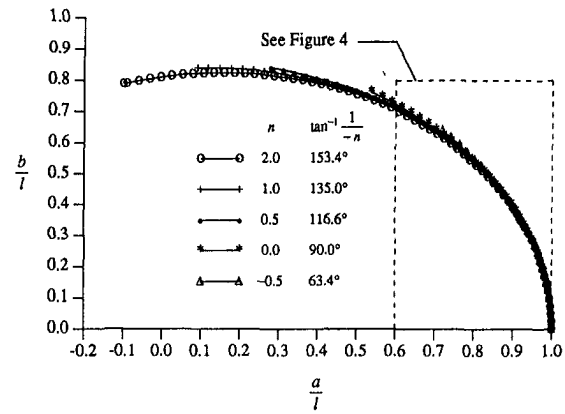

Fig. 3 Beam tip deflection for various end forces

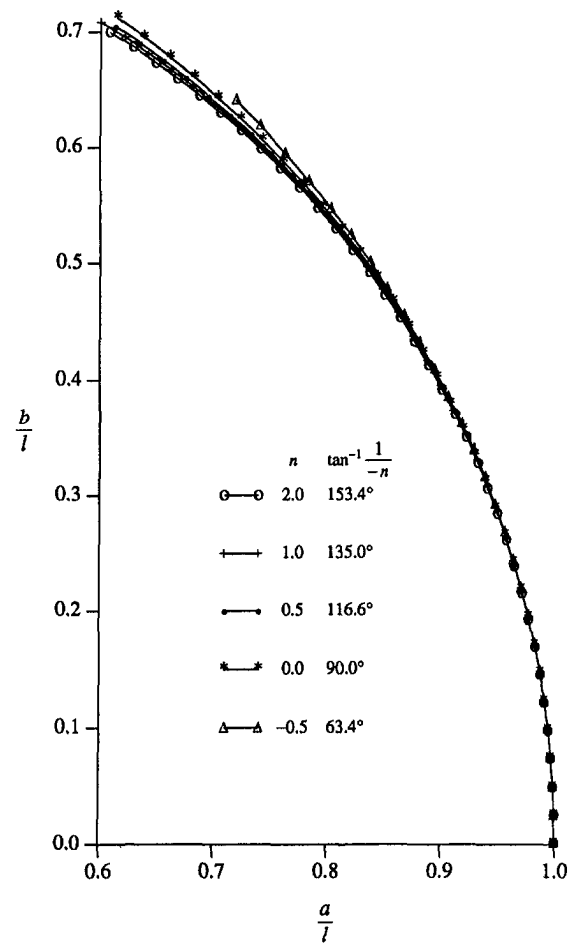

Fig. 4 Beam tip deflection for various end forces

(*ii*) it is noted that these paths are initially nearly circular, with a center of curvature at some point along the length of the undeflected beam. The deflection path for a particular loading diverges from the near circular path as its end angular deflection, θ_o, approaches the angle of the equivalent applied load $\phi = \tan^{-1}(1/-n)$.

These phenomena will be verified and used to obtain simple parametric approximations for the large-deflection analysis of end-loaded cantilever beams.

Pseudo-Rigid-Body Model and Parameterization

The idea, that a large-deflection beam with end loads initially

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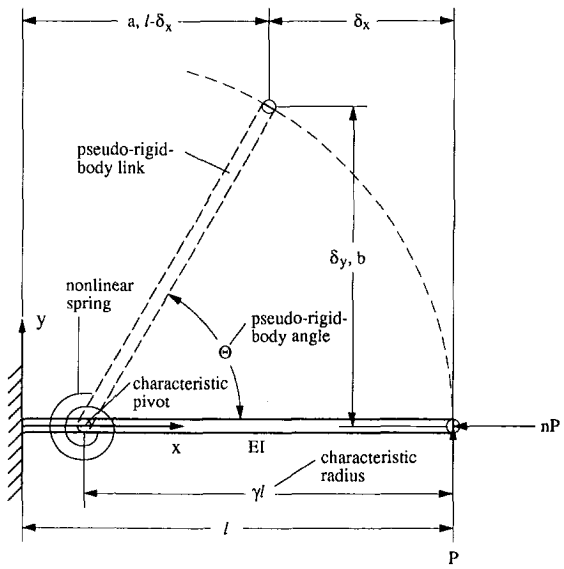

Fig. 5 Parametric approximation of large-deflection beam

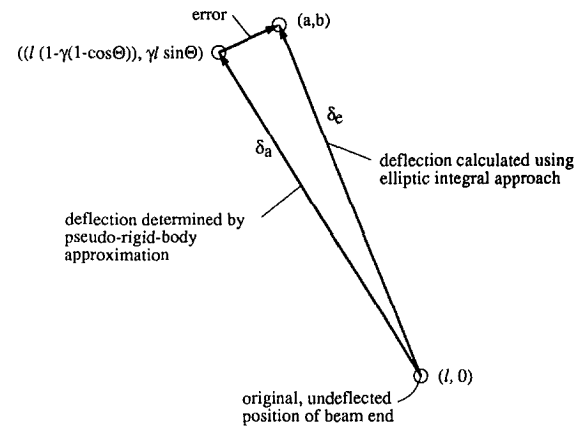

Fig. 6 Determination of error in deflection approximations

follows a near-circular path, will be used to develop parametric approximations for the beam deflection path. A physical model termed a pseudo-rigid-body model will be used to describe these relationships. A pseudo-rigid-body model of a large-deflection beam is shown in Fig. 5. Here, it is assumed that the near circular path can be accurately modeled by two rigid links, joined at a pivot along the beam. A spring with nonlinear stiffness characteristics is placed at the pivot to represent the beam's resistance to deflection. The location of this pseudo-rigid-body "characteristic pivot" is measured from the beam end as a fraction of the beam length, where the fractional distance is γl, and γ is called the "characteristic radius factor." The product γl, the "characteristic radius," is the radius of the circular deflection path traversed by the end of the pseudo-rigid-body link. It is also the length of the pseudo-rigid-body link.

The pseudo-rigid-body approximation described above will be used to parameterize deflection path, beam end angular deflection, and load-deflection relationships in Θ, the "pseudo-rigid-body angle." In equation form:

$$x = f_1(\Theta) \tag{21a}$$

$$y = f_2(\Theta) \tag{21b}$$

$$\theta_o = f_3(\Theta) \tag{21c}$$

$$\alpha^2 = f_4(\Theta) \tag{21d}$$

A discussion on the development of the above relationships follows.

Parametric Approximation of Beam Deflection Path. An acceptable value for the characteristic radius factor, γ, may be found by first determining the maximum acceptable percentage error in deflection. An optimization method is then used to find that γ which would allow the maximum pseudo-rigid-body angle, Θ, while satisfying the maximum error constraint. This one-dimensional optimization problem may be stated formally as follows:

Find the value of the characteristic radius factor, γ, which maximizes the pseudo-rigid-body angle, Θ, where

$$\Theta = \tan^{-1}\left(\frac{b}{a - l\,(1-\gamma)}\right) \tag{22}$$

subject to the parametric constraint

$$g(\Theta) = \frac{error}{\delta_e} \le \left(\frac{error}{\delta_e}\right)_{\max} \quad \text{for } 0 < \Theta < \Theta_{\max} \tag{23}$$

where $error/\delta_e$ is the relative deflection error, and a and b are the horizontal and vertical coordinate of the deflected end, respectively, calculated using the elliptic integral approach. The error in the deflection approximation, as defined here, is illustrated in Fig. 6. The deflections calculated by the elliptic integral approach, δ_e, and by the pseudo-rigid-body approximation, δ_a, are given by

$$\delta_e = \sqrt{(l-a)^2 + b^2} \tag{24}$$

$$\delta_a = \sqrt{(\gamma l(1-\cos\Theta))^2 + (\gamma l \sin\Theta)^2} \tag{25}$$

The error in the deflection is calculated as

$$\frac{error}{l} = \sqrt{\left(\frac{a}{l} - (1-\gamma(1-\cos\Theta))\right)^2 + \left(\frac{b}{l} - \gamma\sin\Theta\right)^2} \tag{26}$$

$$\frac{error}{\delta_e} = \frac{\sqrt{\left(\frac{a}{l} - (1-\gamma(1-\cos\Theta))\right)^2 + \left(\frac{b}{l} - \gamma\sin\Theta\right)^2}}{\sqrt{\left(1-\frac{a}{l}\right)^2 + \left(\frac{b}{l}\right)^2}} \tag{27}$$

One-Dimensional Optimization. Many numerical, one-dimensional optimization methods are available, some of the most common of which are elimination methods, such as the Fibonacci and Golden section methods and interpolation methods, e.g., the cubic and direct root methods (Rao, 1984). The method described and used herein is the Golden Section method. This method is chosen because of its simplicity, the ability to easily specify an interval of uncertainty as the stopping criterion, and the elimination of the need for derivative evaluations.

The Golden Section elimination method is started by choosing an initial interval of uncertainty in which the optimum is known to lie. For instance, consider a function with an interval of uncertainty L_o, where $L_o = x_{right} - x_{left}$, as shown in Fig. 7(*a*). Two experiments are carried out at the points x_1 and x_2, located at a distance of $.382L_o$ from the ends of the interval, i.e., $x_1 = x_{left} + .382L_o$ and $x_2 = x_{right} - .382L_o$, Fig. 7(*a*). Assuming that the function is unimodal, i.e., it has only one local maximum or minimum in the interval, part of the interval may be eliminated. For the maximization problem, the design variable nearest the optimum of a unimodal function will have the highest function value, and the design variable with the smallest

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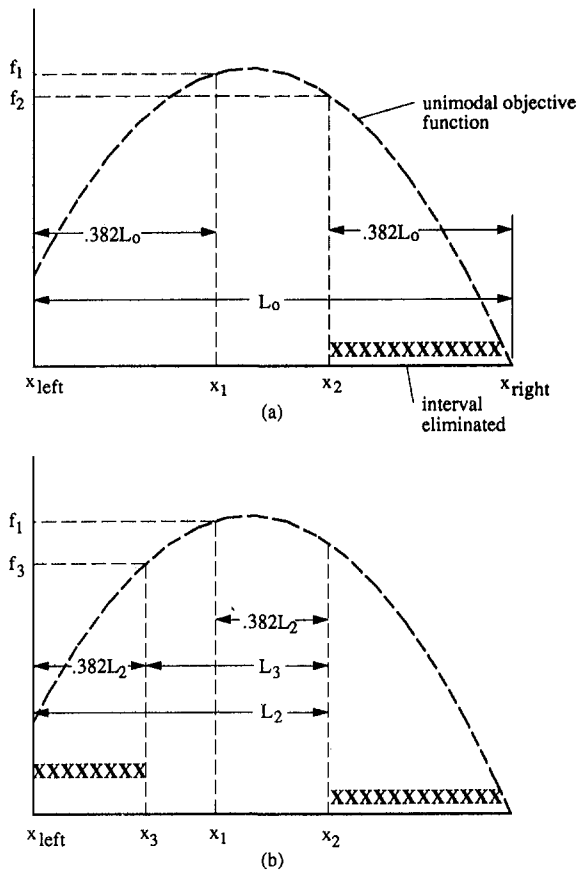

Fig. 7 Golden section method

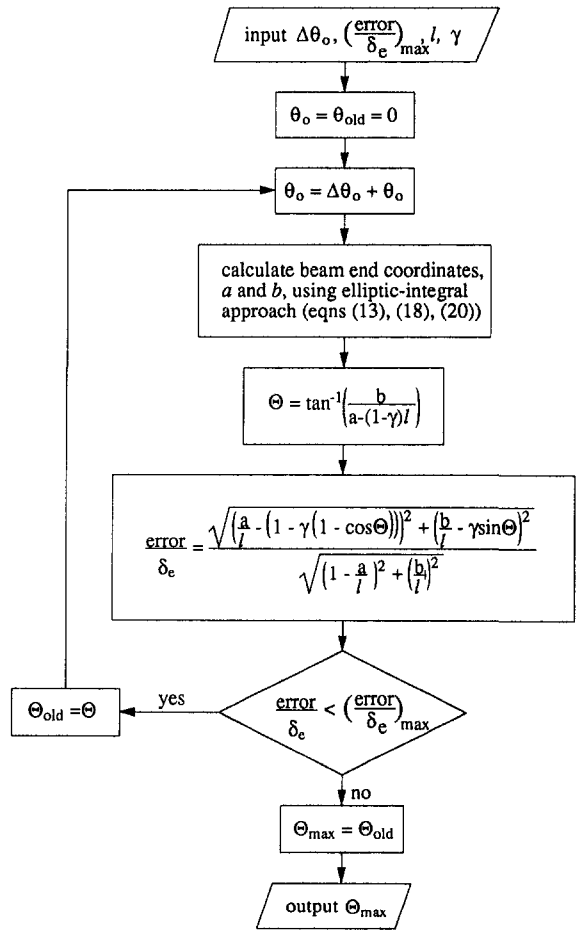

Fig. 8 One-dimensional optimization objective function modified to account for parametric constraint

function value defines one end of the new interval of uncertainty, L_2, Fig. 7(*b*). Two new experiments are now conducted in the new interval of uncertainty at a distance of $.382L_2$ from either end, as shown in Fig. 7(*b*). This method is continued unitl the interval of uncertainty is as small as desired.

The problem at hand is to determine the characteristic radius factor, γ, which will provide the best pseudo-rigid-body estimate for a large-deflection beam. To facilitate the use of the Golden Section method described above, the objective function is modified to take into account the parametric constraint. This modification is shown in a flow chart form in Fig. 8. The maximum relative deflection error, $(error/\delta_e)_{max}$, and the step size for the beam end angular deflection, $\Delta\theta_o$, are required inputs. The beam end angular deflection is initially set equal to $\Delta\theta_o$ and the vertical and horizontal deflections are calculated using the elliptic integral solution of Eqs. (13), (18), and (20). The pseudo-rigid-body angle, Θ, and the relative error in the approximated deflection are determined using Eqs. (22) and (27), respectively. If this error is less than the maximum allowable relative error, $(error/\delta_e)_{max}$, then the beam end angular deflection is incremented by $\Delta\theta_o$ and the process is continued. When the maximum error is exceeded, the previous value of Θ_{max} is taken as the objective function value for the given design variable, γ. The values of $\Delta\theta_o$ and $(error/\delta_e)_{max}$ used in the work that follows are 0.1 deg. and 0.5 percent, respectively.

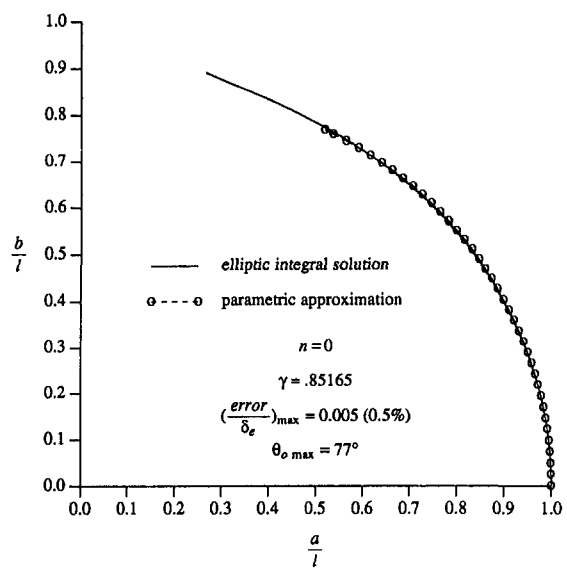

Fig. 9 Parametric approximation of beam deflection path for $n = 0$

Characteristic Radius Factor (γ). The optimization technique described above is used to find the optimal value of the characteristic radius factor, γ. Figure 9 shows the results for $n = 0$, where the optimal γ is found to be .8517. This represents an approximation which is associated with an error smaller than 0.5 percent, a maximum beam end angular deflection of $\theta_{omax} = 77$ deg., and a vertical deflection equal to nearly 80 percent of the beam length. The absolute error, $error/l$, and relative error, $error/\delta_e$, are shown plotted in Fig. 10 with Θ. The value of Θ for zero error represents the point where the approximate and exact solution curves cross.

Figures 11 and 12 show γ and the beam end angular deflection parameterization limit, θ_{omax}, respectively, plotted as functions of n. Equations to describe these relationships will be developed later.

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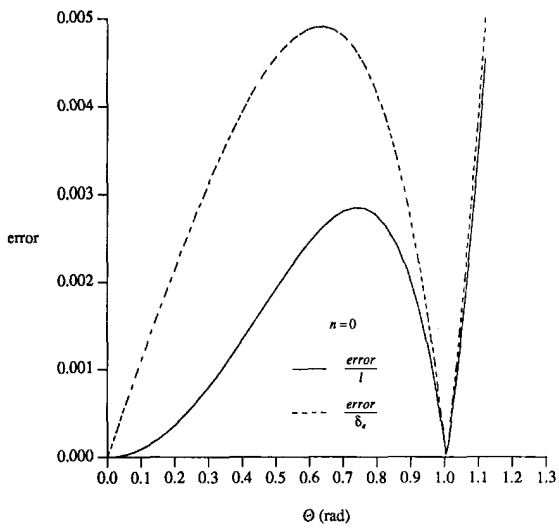

Fig. 10 Error in parametric approximation of deflection for $n=0$

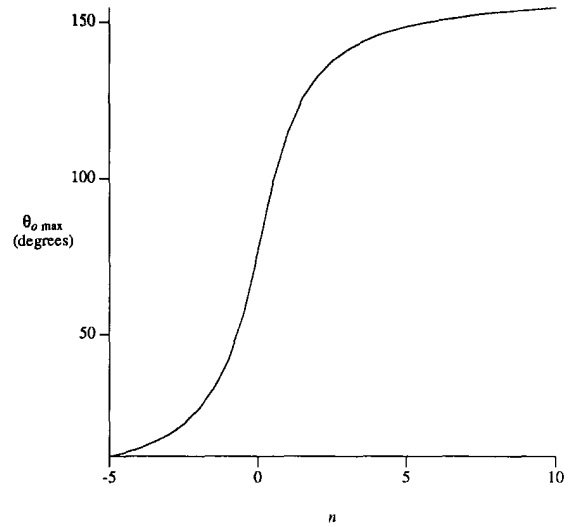

Fig. 12 Beam end angular deflection parameterization limit, θ_{omax} versus n

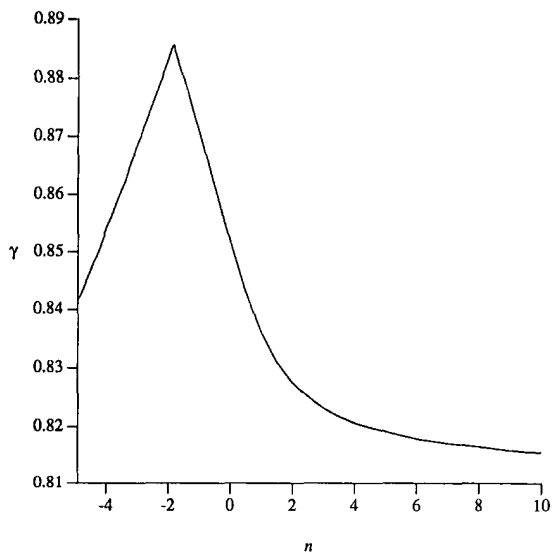

Fig. 11 Plot of characteristic radius factor, γ, versus n

The coordinates of the end of a deflected beam may now be given by equations parameterized in terms of the pseudo-rigid-body angle, Θ (Fig. 5). These equations may be written as

$$\frac{a}{l}=1-\gamma\,(1-\cos\Theta) \tag{28}$$

$$\frac{b}{l}=\gamma\sin\Theta \tag{29}$$

Simplified Relations for γ and θ_{omax}

The values for the characteristic radius factor, γ, plotted in Fig. 11 would be more useful if the value of γ were written as a function of n. Such a relation is possible using linear and polynomial curve-fitting techniques with γ versus n data (Burden and Faires, 1985).

To begin, consider a general nth degree polynomial expressed as

$$y=a_0+a_1x+a_2x^2+\ldots+a_nx^n \tag{30}$$

where, for example, for the problem at hand, $x=n$, and $y=\gamma$. The error in Eq. (30) for the ith data point may be expressed as

$$Error_i=(Y_i-a_0-a_1x_i-a_2x_i^2-\ldots-a_nx_i^n) \tag{31}$$

where Y_i is the exact value of y at data point i. The total error, E, is defined as the sum of the error squarred at all data points:

$$E=\sum_{i=1}^{N}(Error_i)^2$$

$$=\sum_{i=1}^{N}(Y_i-a_0-a_1x_i-a_2x_i^2-\ldots-a_n\,x_i^n)^2 \tag{32}$$

This error is minimized by differentiating Eq. (32) with respect to each coefficient, a_i, i.e.:

$$\frac{\partial E}{\partial a_0}=\sum_{i=1}^{N}2(Y_i-a_0-a_1x_i-a_2x_i^2-\ldots-a_nx_i^n)(-1)\quad=0$$

$$\vdots \qquad \vdots \qquad \vdots \tag{33}$$

$$\frac{\partial E}{\partial a_n}=\sum_{i=1}^{N}2(Y_i-a_o-a_1x_i-a_2x_i^2-\ldots-a_nx_i^n)(-x_i^n)=0$$

Simplifying Eqs. (33) yields:

$$a_0N+a_1\sum_{i=1}^{N}x_i+a_2\sum_{i=1}^{N}x_i^2+\ldots+a_n\sum_{i=1}^{N}x_i^n=\sum_{i=1}^{N}Y_i$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots \tag{34}$$

$$a_0\sum_{i=1}^{N}x_i^n+a_1\sum_{i=1}^{N}x_i^{n+1}+a_2\sum_{i=1}^{N}x_i^n+\ldots+a_n\sum_{i=1}^{N}x_i^{2n}=\sum_{i=1}^{N}Y_ix_i^n$$

The coefficients a_i ($i=0, \ldots, n$) may now be found by solving Eqs. (34) simultaneously.

The correlation coefficient, r, reveals how good the fit is (Holman, 1989). This coefficient is defined in a general form as

$$r=\left(1-\frac{\sigma_{y,x}^2}{\sigma_y^2}\right)^{1/2} \tag{35}$$

where, the standard deviation of y, σ_y, is

$$\sigma_y=\left(\sum_{i=1}^{N}\frac{(y_i-y_m)^2}{N-1}\right)^{1/2} \tag{36}$$

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and, the standard error of estimate of y on x, $\sigma_{y,x}$ is

$$\sigma_{y,x}=\left(\sum_{i=1}^{N}\frac{(y_i-y_{ic})^2}{N-2}\right)^{1/2} \tag{37}$$

and y_i are the actual values of y, y_{ic} the values computed from the least squares equation, and

$$y_m=\frac{1}{N}\sum_{i=1}^{N}y_i \tag{38}$$

Applying the above technique, the following equations are developed:

$$\gamma=.841655-.0067807n+.000438004n^2;\ 0.5<n<10.0 \tag{39a}$$

$$\gamma=.852144-.0182867n;\ -1.8316<n<0.5 \tag{39b}$$

$$\gamma=.912364+.0145928n;\ -5.0<n<-1.8316 \tag{39c}$$

The square of the correlation coefficients (r^2) for the above relationships are .9440, .9996, and .9997, respectively. These represent errors in γ of less than 0.005, or 0.6 percent.

An equation relating the parameterization limit, $\theta_{o\max}$, and n is also needed. This limit represents the beam end angular deflection at which the parametric approximation and the theoretical values begin to diverge. As explained earlier, this divergence occurs as the beam end angular deflection approaches the equivalent load angle, $\phi=\tan^{-1}(1/-n)$. For this reason, it is expected that the limit would be a function of ϕ. Assuming a linear variation, a linear least-squares analysis is performed using $x=\theta_{o\max}$ and $y=\phi$. The result may be used to express $\tan\phi$ as

$$\tan(-.025127+1.1487007\theta_{o\max})=\frac{1}{-n};\ -5.0<n<10.0 \tag{40}$$

The above equation is a close approximation, but it overestimates $\theta_{o\max}$ for some values of n. Such a condition is undesirable since it would suggest the use of parametric approximations at times when, in fact, they are not applicable. For this reason, the linear argument of the tangent function of Eq. (40) is rounded so as to give approximations that have acceptable accuracy but tend to be slightly conservative. This new equation is

$$\theta_{o\max}=0.85\tan^{-1}-\frac{1}{n};\ -5.0<n<10.0 \tag{41}$$

The maximum error in the above equation, for $0.0<n<10.0$, is 6.4 deg. (or 4.3 percent) at $n=10.0$. For $-5.0<n<0.0$, the maximum error is 2 deg. at $n=-5.0$.

Rule of Thumb for γ. A rough approximation of γ may be made by determining its average value over a specified range of n, or

$$\gamma_{av}=\frac{\int_{n_1}^{n_2}\gamma dn}{\int_{n_1}^{n_2}dn} \tag{42}$$

For an equivalent load in the range 135.0 deg. $\le\phi\le$ 63.4 deg. and $-0.5\le n\le 1.0$, γ_{av} is found from Eqs. (39) and (42) as

$$\gamma_{av}=\frac{\int_{-0.5}^{0.5}(.852144-.0182867n)dn+\int_{0.5}^{1.0}(.841655-.0067807n+.000438004n^2)dn}{\int_{-0.5}^{1.0}dn}$$

$$\gamma_{av}\approx .84704$$

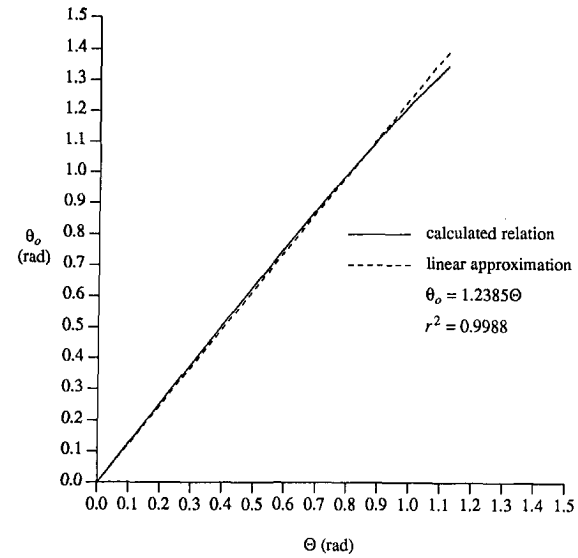

Fig. 13 Linear approximation of θ_o versus Θ for $n=0$

or,

$$\gamma_{av}\approx .85$$

Parametric Approximations of Beam End Angular Deflection

Now that parametric equations have been developed for the deflection path, the next step is to find a relationship between the pseudo-rigid-body angle, Θ, and the beam end angular deflection, θ_o. Recall that the curves of Figs. 3 and 4 have a symbol placed at every 2 deg. of beam end angular deflection. Also note that in the region where the path is nearly circular, the arc lengths along any given path between two consecutive symbols appear to be constant. If the arc length, Δs, is constant between equal increments of θ_o, $\Delta\theta_o$, then a linear relationship is implied and the following may be stated:

$$s=c_s\theta_o \tag{43a}$$

where c_s is a constant of proportionality. Also, from the pseudo-rigid-body model of the beam,

$$s=(\gamma l)\Theta \tag{43b}$$

Equations (43a) and (43b) lead to the linear relationship between θ_o and Θ:

$$\theta_o=c_\theta\Theta \tag{44}$$

where constant c_θ is termed the "parametric angle coefficient." This linear relationship may be tested by performing a linear least squares curve fit and calculating the correlation coefficient to determine the quality of the fit. Figure 13 shows a linear curve fit for a large-deflection beam loaded with a single transverse load ($n=0$). The least squares approximation is based on data taken at every 0.1 deg. of beam end angular deflection up to the limit of approximation accuracy, $\theta_o=77$ deg., and has a correlation coefficient given by $r^2=0.9988$. Table 1 shows this information for various values of n and ϕ, where $\phi=\tan^{-1}(1/n)$.

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Parameterized Load-Deflection Relations

The deflection path and beam end angular deflection are now parameterized and an equation is desired relating the pseudo-rigid-body angle, Θ, and the load index, α^2. The complexity of the relationship requires a polynomial equation to adequately approximate the load-deflection relation. The use of polynomials in such applications is not unusual. For example, the Structural Stability Research Council (SSRC) presents a polynomial curve fit relating the buckling load of a steel column to its slenderness ratio (Chen and Liu, 1987).

The load parameterization relation is found using a 5th degree polynomial, i.e.,

$$\alpha^2 = f(\Theta) = a_1\Theta + a_2\Theta^2 + a_3\Theta^3 + a_4\Theta^4 + a_5\Theta^5 \qquad (45)$$

where a_i, $i = 1, \ldots, N$ are termed the "stiffness coefficients."

Table 1 Parameterized angular deflection data

n	ϕ	c_θ	r^2
0.0	90.0	1.2385	0.9988
0.5	116.6	1.2430	0.9992
1.0	135.0	1.2467	0.9994
1.5	146.3	1.2492	0.9994
2.0	153.4	1.2511	0.9995
3.0	161.6	1.2534	0.9995
4.0	166.0	1.2548	0.9995
5.0	168.7	1.2557	0.9995
7.5	172.4	1.2570	0.9995
10.0	174.3	1.2578	0.9995
-0.5	63.4	1.2348	0.9979
-1.0	45.0	1.2323	0.9963
-1.5	33.7	1.2322	0.9942
-2.0	26.6	1.2293	0.9931
-3.0	18.4	1.2119	0.9939
-4.0	14.0	1.1971	0.9951
-5.0	11.3	1.1788	0.9952

Data points were taken at every 0.1 deg. of beam end angular deflection up to the parameterization limit. The approximations for $n = -0.5$, 0.0, 0.5 and 2.0 are shown in Fig. 14.

In the design and analysis of compliant mechanisms, the desired deflection is often known and the load needed to cause this deflection is calculated, thus rendering Eq. (45) very useful. However, in many applications the load is known and the corresponding deflection is unknown. To remedy this, the resulting polynomial equation takes the form

$$\Theta = b_1\alpha^2 + b_2(\alpha^2)^2 + b_3(\alpha^2)^3 + b_4(\alpha^2)^4 + b_5(\alpha^2)^5 \qquad (46)$$

where b_i, $i = 1, \ldots, N$, are called the "flexibility coefficients."

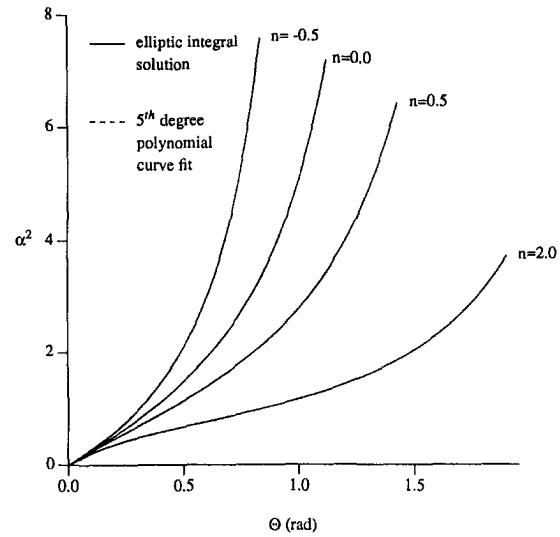

Fig. 14 Plot of α^2 versus Θ for $n = 0.0, -0.5, 0.5, 2.0$

Table 2 Parameterized load-deflection data

n	ϕ	a_1, b_1	a_2, b_2	a_3, b_3	a_4, b_4	a_5, b_5	r^2
0.0	90.0	0.27856E+01	-0.25170E+01	0.10216E+02	-0.12403E+02	0.70012E+01	0.99999
		0.41419E+00	-0.50824E-01	-0.30078E-02	0.13762E-02	-0.93490E-04	0.99997
0.5	116.6	0.26739E+01	-0.24966E+01	0.52299E+01	-0.45257E+01	0.18973E+01	0.99999
		0.41516E+00	0.65483E-01	-0.52909E-01	0.96586E-02	-0.58321E-03	0.99994
1.0	135.0	0.25807E+01	-0.29287E+01	0.41567E+01	-0.28121E+01	0.91374E+00	0.99999
		0.37136E+00	0.32037E+00	-0.18705E+00	0.36960E-01	-0.25465E-02	0.99992
1.5	146.3	0.24781E+01	-0.32796E+01	0.39425E+01	-0.23141E+01	0.62785E+00	0.99999
		0.27518E+00	0.77865E+00	-0.48309E+00	0.11048E+00	-0.89504E-02	0.99987
2.0	153.4	0.23640E+01	-0.34644E+01	0.38583E+01	-0.20887E+01	0.50431E+00	0.99999
		0.12726E+00	0.14978E+01	-0.10455E+01	0.27893E+00	-0.26561E-01	0.99975
3.0	161.6	0.21373E+01	-0.35454E+01	0.37192E+01	-0.18614E+01	0.39516E+00	0.99997
		-0.28984E+00	0.38357E+01	-0.34063E+01	0.11927E+01	-0.14999E+00	0.99927
4.0	166.0	0.19343E+01	-0.34439E+01	0.35372E+01	-0.17087E+01	0.34008E+00	0.99990
		-0.83488E+00	0.75368E+01	-0.82539E+01	0.36139E+01	-0.56975E+00	0.99860
5.0	168.7	0.17602E+01	-0.32801E+01	0.33378E+01	-0.15816E+01	0.30321E+00	0.99976
		-0.14816E+01	0.12761E+02	-0.16712E+02	0.88219E+01	-0.16791E+01	0.99786
7.5	172.4	0.14301E+01	-0.28457E+01	0.28742E+01	-0.13316E+01	0.24355E+00	0.99914
		-0.34488E+01	0.33349E+02	-0.62116E+02	0.47165E+02	-0.12938E+02	0.99600
10.0	174.0	0.12021E+01	-0.24791E+01	0.25008E+01	-0.11476E+01	0.20536E+00	0.99465
		-0.57968E+01	0.65889E+02	-0.15993E+03	0.15920E+03	-0.57304E+02	0.99826
-5.0	11.3	0.40601E+01	-0.96247E+02	0.25194E+04	-0.22038E+05	0.74899E+05	0.99988
		0.36189E+00	-0.46190E+00	0.33721E+00	-0.12211E+00	0.16927E-01	0.99958
-4.0	14.0	0.41534E+01	-0.81566E+02	0.17380E+04	-0.12425E+05	0.34465E+05	0.99988
		0.35609E+00	-0.36431E+00	0.21264E+00	-0.61499E-01	0.68059E-02	0.99957
-3.0	18.4	0.51125E+01	-0.99428E+02	0.14980E+04	-0.80281E+04	0.16257E+05	0.99974
		0.33922E+00	-0.23884E+00	0.91517E-01	-0.16957E-01	0.11868E-02	0.99904
-2.0	26.6	0.59525E+01	-0.92012E+02	0.93345E+03	-0.34600E+04	0.47803E+04	0.99958
		0.32797E+00	-0.15009E+00	0.36320E-01	-0.41905E-02	0.18129E-03	0.99848
-1.5	33.7	0.50054E+01	-0.52377E+02	0.44035E+03	-0.13172E+04	0.14918E+04	0.99975
		0.34256E+00	-0.13589E+00	0.29608E-01	-0.31403E-02	0.12634E-03	0.99929
-1.0	45.0	0.37108E+01	-0.18177E+02	0.13173E+03	-0.30872E+03	0.28733E+03	0.99993
		0.36969E+00	-0.12891E+00	0.26874E-01	-0.28741E-02	0.12018E-03	0.99990
-0.5	63.4	0.30448E+01	-0.53973E+01	0.33634E+02	-0.58216E+02	0.43256E+02	0.99998
		0.39455E+00	-0.10415E+00	0.17426E-01	-0.15976E-02	0.60416E-04	0.99999

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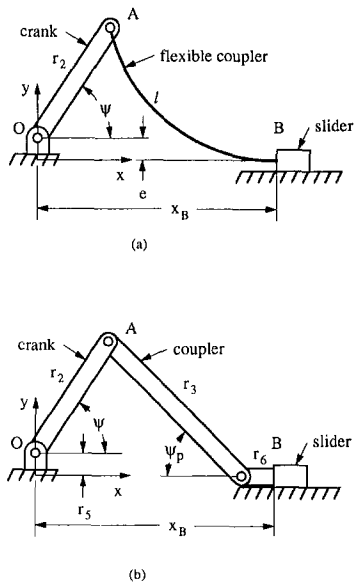

Fig. 15 (*a*) A compliant slider-crank mechanism, and (*b*) its pseudo-rigid-body model

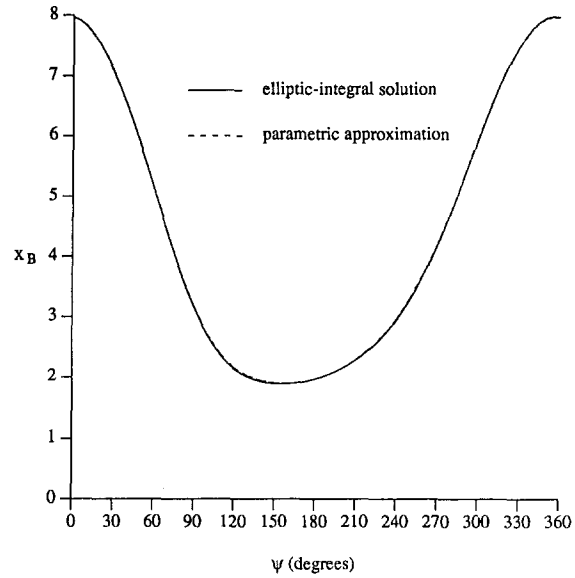

Fig. 16 Slider displacement, X_B, versus Ψ for compliant slider-crank mechanism

Table 2 shows the stiffness and flexibility coefficients in Eqs. (45) and (46) for various values of n, assuming Θ is in radians. The corresponding correlation coefficients, r^2, are also given. For ease of use in computations, the information in Table 2 may be approximated in equation form. This is done by extending the least squares approximation of Eqs. (30) through (38) to two dimensions and performing a bivariate curve fit. This yields:

$$\text{for } 0.0 \le n \le 2.0$$

$$\alpha^2 = 3.080761\Theta - 1.197269\Theta^2 - 1.324163\Theta n + 2.642665\Theta^3 - 0.702815\Theta^2 n + 0.560222\Theta n^2 - 1.615738\Theta^3 n + 0.447698\Theta^3 n^2; \quad r^2 = 0.99560 \tag{47a}$$

$$\text{for } -1.5 \le n \le 0.0$$

$$\alpha^2 = 3.874848\Theta - 4.561288\Theta^2 - 1.244845\Theta n + 5.930629\Theta^3 + 12.131849\Theta_n^2 - 0.251214\Theta n^2 - 15.288859\Theta^3 n + 22.432280\Theta^3 n^2; \quad r^2 = 0.99728 \tag{47b}$$

Example

Consider the compliant slider-crank mechanism shown in Fig. 15(*a*), where a flexible segment length, l, of 5.0 units, a crank length, r_2, of 3.0 units, and an offset, e, of 0.5 units are used. Plot the slider displacement, x_B, and the required crank input torque, T_{in}, with the crank input angle, α, using the parametric approximations developed herein. Assume the following: the slider always remains in contact with ground, there is no frictional resistance to slider motion, and the flexible segment is initially straight. Compare the results to that obtained by the elliptic integral approach.

Since the slider will not support a force in the horizontal direction, the force on the end of the flexible coupler is necessarily vertical, i.e., $n = 0$. Using Eqs. (39*b*) and (41), γ and $\theta_{o\max}$ are calculated to be 0.852 and 76.5 deg., respectively. The load relations may be determined from Eq. (47) or Table 2. A pseudo-rigid-body model is developed as shown in Fig. 15(*b*). This model may now be analyzed using classical rigid-body mechanism equations (Paul, 1979):

$$r_3 = \gamma l \tag{48}$$

$$r_6 = l - r_3 \tag{49}$$

$$y_A = r_2 \sin \Psi + r_5 \tag{50}$$

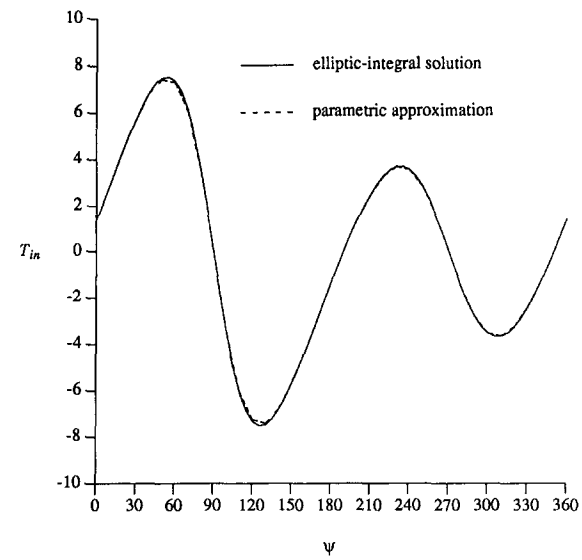

Fig. 17 Input torque requirements for compliant slider mechanism

$$\Psi_p = \sin^{-1}\left(\frac{|y_A|}{r_3}\right) \tag{51}$$

$$x_B = r_2 \cos \Psi + r_3 \cos \Psi_p + r_6 \tag{52}$$

$$T_{in} = P\, r_2 \cos \Psi \tag{53}$$

Figure 16 shows the slider coordinate, x_B, versus Ψ, calculated using the above equations and compared to the solution found from the use of the closed-form elliptic integral equations. Results for the input torque are shown in a similar manner in Fig. 17.

Conclusion

The geometric nonlinearity associated with large-deflection beams makes their analysis very complicated. Developed herein is a method of approximating the deflection path of end-loaded large-deflection cantilever beams. A physical model associated with this approximation allows simplification of complex prob-

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lems by means of a pseudo-rigid-body model. The values for γ, a, b, and θ_{omax} presented herein have been found from deflections calculated by use of the elliptic integral equations derived earlier, and have been verified by a numerical finite element "chain" algorithm (Harrison, 1979; Her, 1986; Howell and Midha, 1992).

Several applications have been identified where implementing the above developments may be useful. Some of these are: (*i*) The kinematic analysis and design of compliant mechanisms: The pseudo-rigid-body model of a flexible beam may be used to model flexible links of compliant mechanisms. This would not only be valuable in determining the motion of compliant mechanisms, but would also be a very useful tool in the design of such mechanisms to perform specific tasks. (*ii*) An approximation used to help iterative numerical solution techniques converge: Some numerical methods and elliptic integral solutions require a shooting method when boundary conditions must be satisfied (Winter and Shoup, 1972; Harrison, 1973; Her, 1986). Convergence difficulties may be minimized or eliminated by selection of good initial estimates for the unknown design variables. The parametric approximations may be used as a means of obtaining very accurate initial estimates. (*iii*) The analysis and design of large-deflection structural components: Though originally developed for use in compliant mechanism analysis and design, the parametric approximations may be effectively used in other application areas where large deflections are encountered.

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